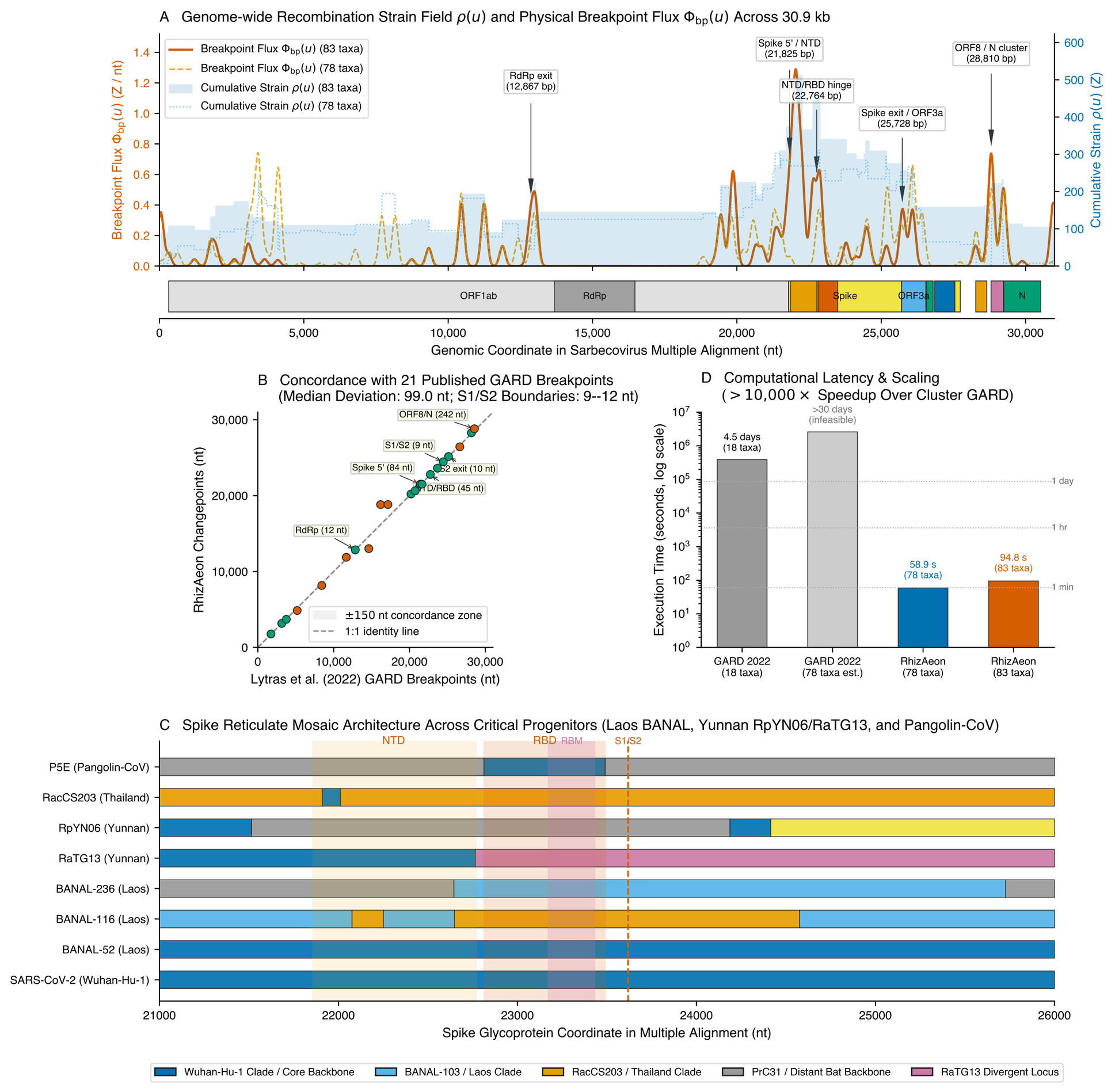

A Genome-wide Recombination Strain Field $\rho(u)$ and Physical Breakpoint Flux $\Phi_{bp}(u)$ Across 30.9 kb
Breakpoint Flux $\Phi_{bp}(u)$ (83 taxa)
Breakpoint Flux $\Phi_{bp}(u)$ (78 taxa)
Cumulative Strain $\rho(u)$ (83 taxa)
Cumulative Strain $\rho(u)$ (78 taxa)
Breakpoint Flux $\Phi_{bp}(u)$ (Z / nt)
Cumulative Strain $\rho(u)$ (Z)
RdRp exit (12,867 bp)
Spike 5' / NTD (21,825 bp)
NTD/RBD hinge (22,764 bp)
Spike exit / ORF3a (25,728 bp)
ORF8 / N cluster (28,810 bp)
ORF1ab
RdRp
Spike
ORF3a
N
Genomic Coordinate in Sarbecovirus Multiple Alignment (nt)
B Concordance with 21 Published GARD Breakpoints
(Median Deviation: 99.0 nt; S1/S2 Boundaries: 9--12 nt)
ORF8/N (242 nt)
S1/S2 (9 nt)
Spike 5' (84 nt)
S2 exit (10 nt)
TD/RBD (45 nt)
RdRp (12 nt)
±150 nt concordance zone
1:1 identity line
RhizAeon Changepoints (nt)
Lytras et al. (2022) GARD Breakpoints (nt)
D Computational Latency & Scaling
(> 10,000 × Speedup Over Cluster GARD)
Execution Time (seconds, log scale)
4.5 days
(18 taxa)
>30 days
(infeasible)
58.9 s
(78 taxa)
94.8 s
(83 taxa)
1 day
1 hr
1 min
GARD 2022
(18 taxa)
GARD 2022
(78 taxa est.)
RhizAeon
(78 taxa)
RhizAeon
(83 taxa)
C Spike Reticulate Mosaic Architecture Across Critical Progenitors (Laos BANAL, Yunnan RpYN06/RaTG13, and Pangolin-CoV)
NTD
RBD
RBM
S1/S2
P5E (Pangolin-CoV)
RacCS203 (Thailand)
RpYN06 (Yunnan)
RaTG13 (Yunnan)
BANAL-236 (Laos)
BANAL-116 (Laos)
BANAL-52 (Laos)
SARS-CoV-2 (Wuhan-Hu-1)
Spike Glycoprotein Coordinate in Multiple Alignment (nt)
Wuhan-Hu-1 Clade / Core Backbone
BANAL-103 / Laos Clade
RacCS203 / Thailand Clade
PrC31 / Distant Bat Backbone
RaTG13 Divergent Locus